QUANTUM CAMPAIGN TARGETING

Who to call, and how likely the campaign is to work

A hybrid quantum-classical pipeline for a bank term-deposit campaign

The Atomiq Hackathon, Qiskit Fall Fest 2026 at IEM | Sidharth

SIX CUSTOMERS TO CONTACT

C0031, C0079, C0081, C0318, C0005, C0170

0.3382

P(at least 3 subscribe), quantum estimate, 1,048,576 shots

0.3385

exact value from the same probabilities

0.0

gap to the brute-force optimum (objective 24.53)

ARCHITECTURE

One pipeline, three linked stages

Stage 1 Predict

300 labelled customers
100 test probabilities
top 10 move on

QUANTUM

6-qubit ZZ feature map,
fidelity quantum kernel

CLASSICAL

preprocessing, SVM, Platt scaling,
logistic regression, 50/50 average

→

Stage 2 Select

exactly 6 of 10
max expected net value
budget and capacity limits

QUANTUM

QAOA on a 10-qubit QUBO,
two variants, sampled on Aer

CLASSICAL

QUBO and Ising map, COBYLA,
brute-force reference, decoding

→

Stage 3 Simulate

P(at least 3 of 6 subscribe)
one qubit per customer

QUANTUM

six RY rotations, measured;
coherent counter circuit

CLASSICAL

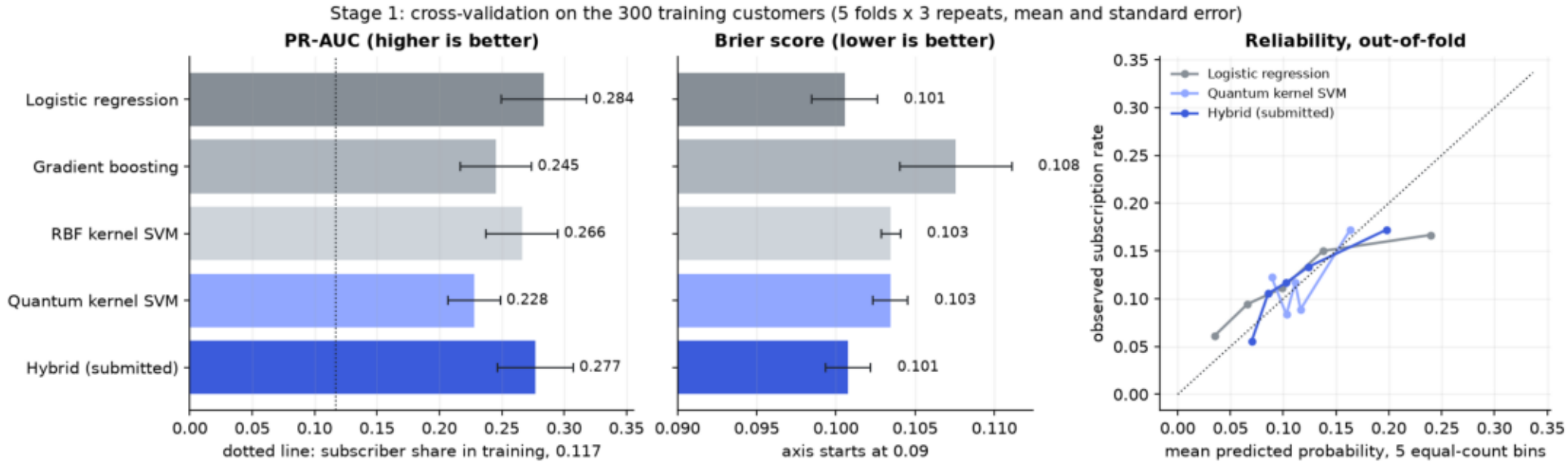
exact Poisson-binomial,
Monte Carlo, readout inversion

The same customer ids and the same Stage 1 probabilities run through all three result files. A traceability check reads the files back and stops the run if anything differs.

PREDICT

Stage 1: a hybrid classifier, validated on training data only

- 300 training customers, 35 subscribers. 5-fold cross-validation, 3 repeats, same folds for every model. The test rows are predicted once.
- Quantum kernel classifier: 6 qubits, ZZ feature map, SVM, Platt scaling. Alone it reaches PR-AUC 0.228, logistic regression 0.284, a classical RBF kernel on the same inputs 0.266.
- Submitted: the fixed 50/50 hybrid, by design. PR-AUC 0.277, Brier 0.1008, log loss 0.3508, calibration slope 0.88.
- No quantum gain here, and I do not claim one. The differences between models are within the fold-to-fold noise.



SELECT

Stage 2: six of ten, written as a QUBO

```
maximise    sum v_i x_i        v_i = p_i * deposit_i - cost_i
subject to  sum x_i = 6
            sum cost_i x_i     <= 25
            sum capacity_i x_i <= 22

E(x) = - sum v_i x_i + A (sum x_i - 6)^2
       + [ -l1 g(x) + l2 g(x)^2 ]  for budget and capacity,
       g(x) = limit - sum w_i x_i
```

- Unbalanced penalties: 10 qubits, no auxiliary qubits. Weights chosen by a stated rule and checked on all 1,024 bitstrings.
- Slack-variable version built and verified too: 18 qubits, exact, 137 couplings against 45.
- Ising map x = (1 - z) / 2, energies equal on every state.

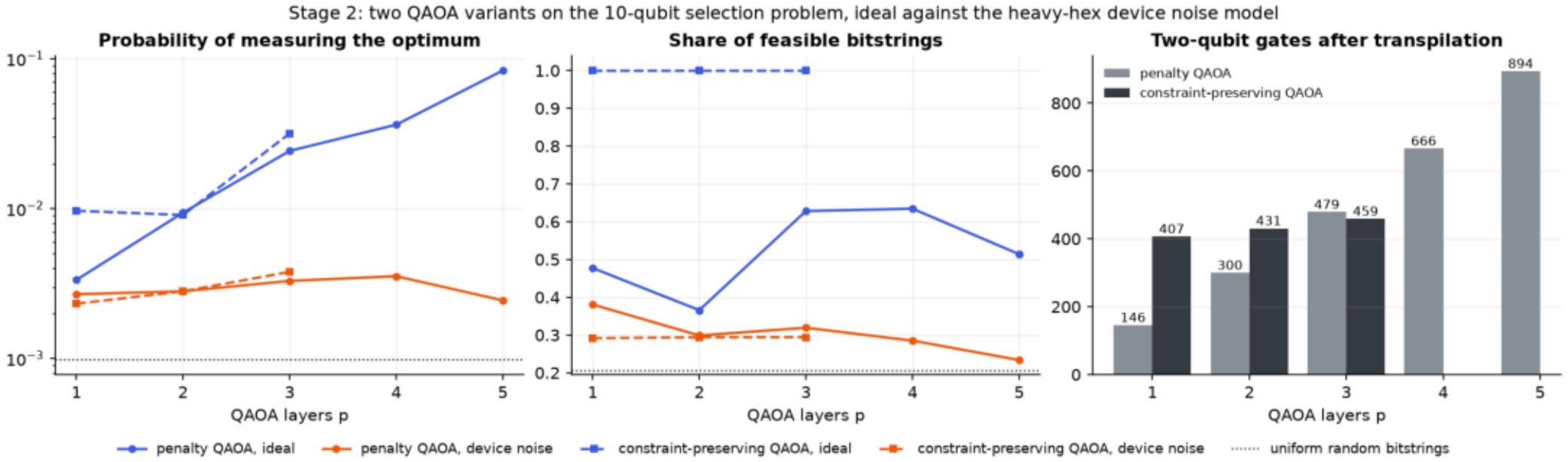
THE LIMITS NEVER BIND FOR THESE TEN CANDIDATES

The costliest group of six costs 20 (limit 25) and the most capacity any group needs is 17 (limit 22). All 210 groups are feasible. I kept the full constraint handling and added a stress instance with tightened limits (5 feasible groups, a different optimum) to test it where it matters.

SELECT

Stage 2: what QAOA adds, measured per shot

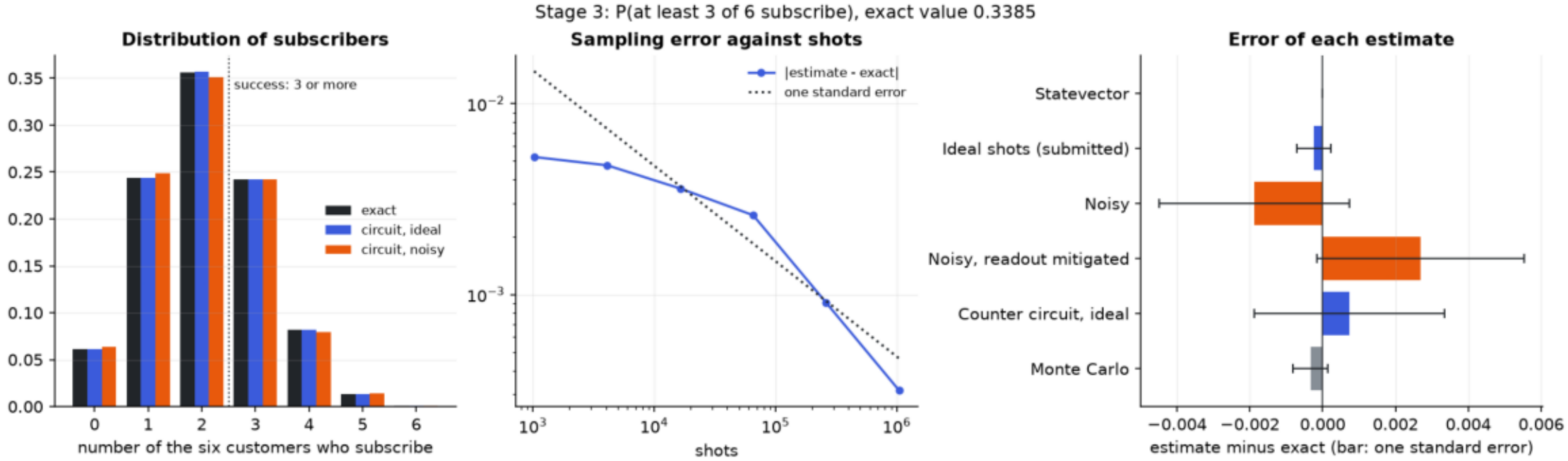
- Penalty QAOA, p = 3: 2.4% of the probability on the optimum, 25 times a random bitstring and 5.1 times a random valid selection of six. Constraint-preserving QAOA (Dicke state + XY mixer): 3.2%, every bitstring has exactly six ones.
- The best feasible measured bitstring is the brute-force optimum, gap 0.0. Both variants agree.
- Honest reference: with 8192 shots, random bitstrings contain the optimum 99.97% of the time. Finding it is not the achievement at 10 qubits.
- Device noise model, heavy-hex: 479 two-qubit gates at p = 3 and the optimum probability falls to 0.0033.



SIMULATE

Stage 3: six qubits, one question

- Qubit i is the i-th selected customer in Stage 1 rank order. RY(2 asin(sqrt(p_i))) makes P(1) equal that customer's probability.
- Success: at least three qubits read 1. Bit order is checked through the per-customer marginals, not only the count.
- Submitted quantum estimate 0.338246 against exact 0.338495 (error -0.000249, one standard error 0.000462).
- A coherent counter circuit (adder + comparator, 13 qubits) gives the same answer on one flag qubit, the starting point for amplitude estimation. It does not survive the noise model.



NOISE AND ERROR PROPAGATION

Where the error comes from

0.0005

shot noise of the submitted Stage 3 estimate

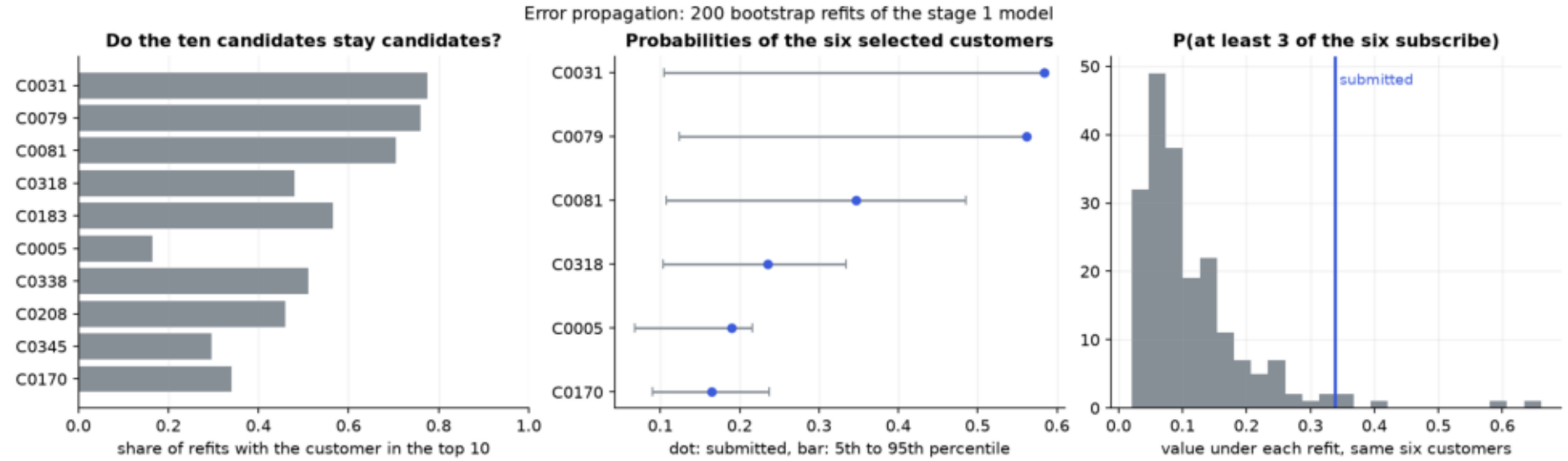
0.0039

readout bias on the Stage 3 circuit (device noise model)

0.088

Stage 1 model uncertainty, 200 bootstrap refits

Stage 1 dominates by a factor of about 191. On average 5.1 of the ten candidates stay in the top 10 of a refit, and the same six customers get a lower probability in 98% of the refits. The answer is very sensitive to the training sample.



CLASSICAL VS QUANTUM

Every quantum result has a classical check next to it

CHECKED

- Quantum kernel against logistic regression, boosting and an RBF kernel on the same inputs and folds
- Kernel entries from the overlap circuit against exact values (mean difference 0.0016)
- QUBO ground state and QAOA selection against brute force
- QAOA sampling against uniform random bitstrings
- Stage 3 circuit against the exact formula, enumeration and Monte Carlo
- Training emulators against Qiskit state vectors of the real circuits

NOT CLAIMED

- No quantum advantage, in accuracy or speed, at any stage. Brute force needs 1,024 evaluations.
- QAOA finding the optimum is not evidence of quality at this size. The per-shot probability is.
- The Stage 3 circuit samples a distribution a laptop samples just as well. Its error falls like Monte Carlo's.
- No real hardware. The noise model is a stated assumption, not a device calibration.

BEYOND THE BRIEF

Three questions the brief does not ask

15% to 31%

model's chance that the campaign loses money when the six customers are correlated (factor parameter 0 to 0.5)

6.5x

smaller error with amplitude estimation at 8,512 state preparations (ideal simulator only)

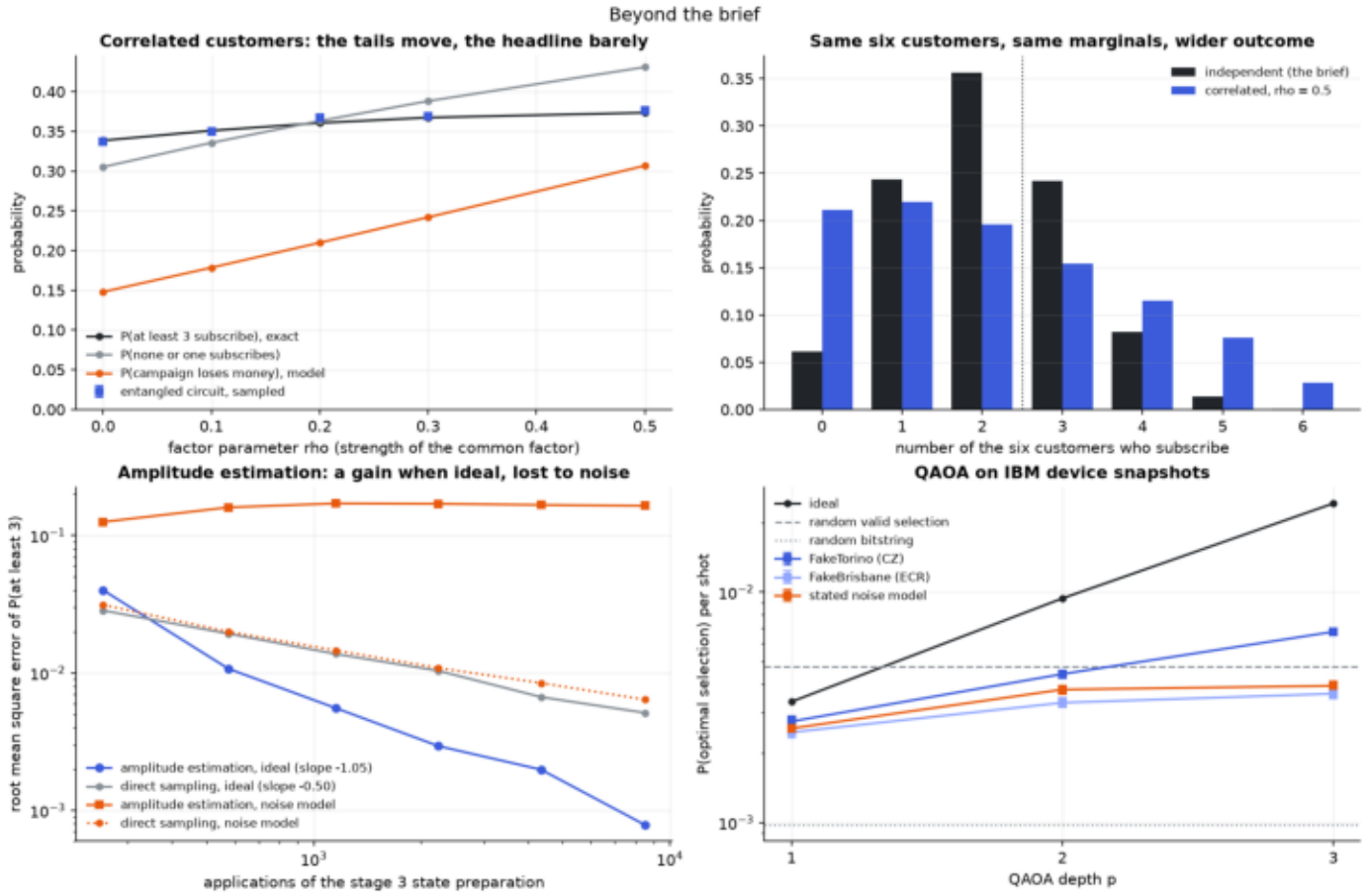
0.67%

of shots return the optimum at p = 3 on the IBM Torino snapshot (random valid selection 0.48%, ideal 2.43%)

- Not independent? A one-factor correlation model on an entangled 9-qubit circuit keeps every Stage 1 probability. P(at least 3) barely moves (0.338 to 0.374) and the expected net value not at all, while the model's loss probability doubles (payoff = expected deposit value).
- Fewer circuit runs? On an ideal simulator the error of amplitude estimation falls with slope -1.05 (direct sampling -0.50). On my noise model it fails: error 0.16 against 0.006. One Grover step already needs 214 two-qubit gates.
- Real machines? On IBM Torino and Brisbane calibration data the Stage 3 estimate is within its standard error after readout mitigation. For QAOA, Torino is 1.7 times better than my stated noise model, and Brisbane matches it.

Same answer, a wider picture around it

- Every panel is checked against exact maths, rebuilt by python run_extensions.py, and none of it enters the three stage files.



BUSINESS IMPACT

What the bank should do

- Contact C0031, C0079, C0081, C0318, C0005, C0170. Expected net value 24.53, cost 15 of 25, capacity 12 of 22.
- Do not plan on three subscriptions. The model says 34%, with 2.1 subscribers expected, and refits of the model on resampled data usually give the same six less: 3% to 26%.
- The limits are not the bottleneck, the exactly-six rule is: 10 units of budget and 10 of capacity stay unused. Without that rule the best campaign inside the same limits contacts 9 of the ten for 27.20.
- The cheapest improvement is more labelled data. The whole pipeline rests on 35 past subscribers, and that uncertainty is far larger than anything the quantum steps add.
- At 10 candidates a laptop solves the selection exactly. What the quantum part gives today is a formulation verified at small size, not a speed-up.

Reproduce it

```
pip install -r requirements.txt
python run_pipeline.py     # every result file and figure, a few minutes on a laptop
python run_extensions.py   # the beyond-the-brief studies, under two minutes
python -m pytest           # the test suite, including a byte-for-byte rerun
```

- Python 3.12, Qiskit 2.6.2, Qiskit Aer 0.18.2, scikit-learn 1.9.1. Versions pinned, every seed in one config file.
- Tests cover leakage, the tie-break rule, file schemas, id and probability traceability, QUBO against brute force, Ising against QUBO, emulators against Qiskit, bit order, the Stage 3 maths and readout mitigation.
- The README is filled in from the result files by a script, so the report cannot drift from the data.

github.com/sidharth-n/quantum-campaign-targeting